Oscar Byrne

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Motivation

I'm a curious QA developer who aspires to keep test theory at the cutting edge. I love getting to ask the gritty questions, like: when is a system considered 'tested', and is automation always the best solution? When writing code, my mantra: test code can and should be the most beautiful in the project. More bugs are lurking in our tests than we'd like to admit!

Work Experience

Cisco, Software Engineer

January 2016 - July 2016

Acano was acquired by Cisco for $700m, after which I was moved to a new agile team to develop a next-gen, **Android-based** product running on custom hardware. This was a **greenfield project**, and from the beginning I wanted our test strategy to be the best it could possibly be.

Accomplishments:

- Ensured build quality by developing a continuous-integration environment using technologies including: **Jenkins**, **Phabricator** and **Docker**.
- Optimised development cycle with a custom **test runner** able to run tests in parallel, and collect useful reports including screenshots. This involved hacking in **Java**, **Gradle**, and **Python**.
- Improved test validity by constructing an all-**hardware test lab**, including network provisions and remote power switching (there was a risk that early prototypes would burst into flames if left on overnight!).
- Encouraged a move towards behaviour-driven development (BDD) by developing a **Java framework** for writing concise, human-readable tests.

Acano, QA Engineer

October 2014 – January 2016

I joined the QA team at Acano in the first wave of graduate recruits. Proving myself with manual testing and writing test plans, I soon graduated to developing frameworks for **automated tests**.

Accomplishments:

- Greatly improved our manual test coverage by rewriting much of the **test plan**. This had a noticeable effect on the stability of our product.
- Helped to quickly deliver a very technical component (**SIP to H323** bridge) by working closely with a single developer.
- Drastically reduced the development cycle for our client software by championing test automation. I designed and implemented the automation framework, written in **Python** and **shell**, from the ground up to be usable by testers with minimal programming experience. This involved rewriting existing core systems to accommodate the new module and implementing a 'write once run anywhere' system able to run tests on platforms including: **Android, iOS, Windows, Mac** and various **web browsers**.

DESY, Summer Studentship Computing Program

July 2013 – August 2013

A 2 month paid internship working with the world's **most brilliant X-ray source** at DESY (Germany's answer to CERN). Aside from a minor project involving RPC technologies, I cut my teeth on Python developing an automated toolchain for processing tomographic images.

Hobbyist Portfolio

Arduino-Based Synthesiser

https://github.com/oscarbyrne/grains

I became interested in the Arduino platform for generating audio after building some simple audio-based electronic circuits. This has involved developing my own **1-bit DAC**, requiring a **custom PWM** implementation, which I ultimately intend to output audio generated on the chip. Getting this close to the metal was a great experience, and I really enjoyed pushing the limits of the chip.

Python Meta-Programming

https://github.com/oscarbyrne/notes
https://github.com/oscarbyrne/vikings

I am interested in developing beautiful interfaces, to the point where they almost become programming languages in their own right. Python's high capacity for metaprogramming makes this possible. I have **2 projects** available on my github demonstrating this - 'notes', which is a tool for exploring music theory; and 'vikings', the latest iteration of a modern 'text adventure' game I have been toying with for several years (note this is very much a work in progress).

This CV

https://github.com/oscarbyrne/cv

I have written this CV with the help of the **reStructuredText syntax**. This is a markup language designed to compile easily to PDF and HTML, whilst remaining easily human-readable. I honestly think my CV is the best (looking) out there written using this language!

Education

University of Birmingham, MSci Physics

2010 - 2014

For my fourth year project I became involved with **front-line research** in metamaterials, and contributed to a project which was presented as part of the PIERS conference 2014 in China. I chose mostly computing-related modules, including:

- **Computational Modelling of Physical Systems,** for which I implemented a dynamical billiards simulation using **C++**
- **Teaching in Schools,** for which I spent 11 weeks teaching in a Primary school
- **Image Processing,** which mostly focused on algorithms

Sir Thomas Rich's School, Gloucester

A levels:	A*, A, B, C
AS levels:	B
GCSEs:	4 A* passes, 5 As and 2 Bs

Other Interests

- Producing electronic music - I've assembled a **home studio** which I use to make hip hop
- Rock climbing

References

References are available on request